



TRANSITION OF CREATIVE AND COGNITIVE PARADIGMS FROM ANTHROPOCENTRIC TO ALGORITHMIC THINKING IN CONTEMPORARY JOURNALISM

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Abstract. This thesis analyzes how creative and cognitive paradigms in modern journalism are evolving during the transition from anthropocentric thinking to algorithmic thinking. It examines how generative artificial intelligence (AI) systems reshape journalistic creativity, authorship, and epistemology by transferring certain cognitive functions from humans to machines. Drawing on process philosophy, posthumanist theory, and media ontology, the study conceptualizes algorithmic journalism as a hybrid form of cognition that integrates human intentionality with machine optimization. It further discusses how this paradigm shift challenges the notions of originality, credibility, and ethical responsibility in the newsroom. The findings suggest that journalism is entering an era of distributed creativity and hybrid authorship, where the role of the journalist evolves from individual storyteller to collaborative curator of human-machine intelligence.

Keywords: algorithmic journalism, creativity, authorship, posthumanism, hybrid cognition, artificial intelligence, epistemology of media



Introduction

Creativity and a conscious human touch have long been central to journalism's identity. Traditionally, reporting relies on human creativity, intuition, and subjective judgment—an anthropocentric cognition that places the journalist's mind at the core of news production. Today, however, the field is witnessing an epistemic shift as algorithmic logic enters the scene. Advances in generative AI and automation are transforming how news is created, moving some creative and cognitive tasks from humans to machines. The rise of large AI models, such as OpenAI's ChatGPT released in November 2022, demonstrated an uncanny ability to generate text and images with human-like proficiency, taking on roles of creation once considered uniquely human (Dwivedi et al., 2023; Mesko, 2023; Roumeliotis et al., 2023). In journalism, this marks a pivotal change: AI now directly intervenes in core creative processes, challenging long-held norms of authorship, originality, and professional identity (Lewis, 2025). Globally, news organizations are experimenting with automation—from robot reporters to AI-edited images—thereby redefining what it means to be an author or storyteller in media (Associated Press, 2015; Thomson et al., 2024). The aim of this discussion is to conceptualize this transition from human-centered to algorithm-driven journalism and to identify its theoretical implications. In short, how does journalism evolve when creativity and “consciousness” in news-making migrate from the journalist's mind to the algorithm's logic?

Result and Discussion

Process philosophy offers one useful lens for interpreting the transformation of creativity in journalism. Alfred North Whitehead famously described creativity as “the category of the ultimate,” a fundamental principle underlying all existence (Whitehead, 1929/1978). In his metaphysical system, creativity is “the universal of universals characterizing ultimate matter of fact,” the principle by which “the many become one, and are increased by one” (Internet Encyclopedia of Philosophy, n.d.; Whitehead Research Encyclopedia, n.d.). In this view, every act of becoming or



knowing is grounded in creativity. Applied to journalism, this highlights that the creative act of story-making is not a trivial addition but the very essence of the practice. News is not merely a compilation of facts; it is created knowledge shaped by the journalist's imaginative and interpretive efforts. If creativity is the ultimate category in human cognition, then replacing or augmenting it with algorithmic systems is no small matter—it suggests a shift at the deepest ontological level of how journalistic content comes into being.

Source Credibility And The Ontology Of Authorship

Source credibility and the ontology of authorship represent an important site of theoretical inquiry, especially as algorithmic creation challenges established assumptions about who or what “authors” content. According to Source Credibility Theory, a message's impact depends heavily on perceptions of the source's expertise and trustworthiness (Jia, 2021). Traditionally, journalistic credibility arises from a reporter's perceived professional knowledge, integrity, and accountability. But when the “source” is an algorithm, audiences may struggle to judge credibility, since an AI lacks a human reputation or moral accountability. Some studies find that human-authored content is rated slightly more credible than algorithmic content (Waddell, cited in Jia, 2021), though results are mixed (Jia, 2021). From a media-ontological lens, the notion of authorship itself is being redefined. Michel Foucault's author-function suggests that “author” is not a fixed individual but a role or discursive function that shapes how texts are interpreted (Shtoltz, 2024). In this framework, some scholars propose the concept of the “algorithmic author” — that large language models and news bots should be understood as participants in authorship, not merely tools (Shtoltz, 2024). Thus, in algorithmic journalism the “author” becomes a composite of human intent and machine generation — a hybrid node in the network of content production. This challenges traditional ontology of media creation: when an AI-written article is published, is the “author” the algorithm, the programmer, the



editor, or all of them? Such questions push journalism to reconceptualize who (or what) speaks in AI-generated news.

AI's Challenge To Creative Cognition

AI's challenge to creative cognition becomes more evident when we bring these ideas together. If creativity is fundamental (as in process philosophy) and we accept hybrid or posthuman agency (Hayles, Braidotti), then the cognitive act of journalism is no longer purely human. Algorithms today can perform selection, pattern recognition, and style mimicry — tasks once reserved for human minds. Generative AI may not “understand” meaning, but it can assemble facts and prose that appear coherent and even creative (Huschens, Briesch, Sobania, Rothlauf, & others, 2023). This undermines the anthropocentric assumption that genuine creativity must rest on human consciousness or subjectivity. Yet posthumanist theory would argue these algorithmic agents are not intruders, but part of an evolving assemblage of human + machine cognition, compelling us to expand our conception of journalistic thought.

Key Oppositions In The Shift From Anthropocentric To Algorithmic Thinking

The transition from human-centered to algorithmic journalism can be illuminated through three conceptual tensions: (1) human intention versus machine optimization, (2) subjective experience versus data-driven neutrality, and (3) creative process versus predictive modeling. Each reveals how journalistic cognition and creativity are being re-engineered by artificial intelligence.

1. *Human Intention vs. Machine Optimization.* Journalists operate with conscious *niyat*—a purposive awareness guiding how stories are selected, framed, and narrated to inform or move the public. This intentionality gives journalism moral direction and contextual depth. Algorithms, by contrast, lack intent; they optimize outputs according to statistical probability or engagement metrics. Generative systems such as OpenAI's ChatGPT generate text by maximizing the likelihood of the next word rather than deciding why a topic matters. The contrast is clear in the



Los Angeles Times "Quakebot," which automatically produced an earthquake report within minutes by filling a template with seismic data (Associated Press, 2015). The code executed flawlessly but without editorial judgment. As automation spreads, journalism risks devolving into optimized outputs detached from human purpose. Sustaining human intentionality—why this story matters—remains essential to preserve ethical and narrative integrity.

2. *Subjective Experience vs. Data-Driven Neutrality.* Journalistic practice has long balanced subjectivity and objectivity. Human reporters draw on lived experience and empathy to interpret events, whereas AI systems process information computationally, appearing "neutral." Yet this neutrality is deceptive: algorithms reproduce biases embedded in data and design. Generative models tend to favor what is statistically common, thus "trending toward the generic" in storytelling. What seems impartial may in fact reflect the dominant patterns of training data—an algorithmic subjectivity. Without experiential awareness, AI risks homogenizing news and narrowing its representational range. A hybrid approach—using algorithms for data analysis while relying on journalists for contextual and ethical interpretation—can mitigate this imbalance.

3. *Creative Process vs. Predictive Modeling.* Human creativity in journalism is nonlinear and exploratory; it thrives on intuition, cultural memory, and occasional rule-breaking. AI's creativity, however, is predictive: it recombines existing patterns to generate statistically plausible outputs. Scholars argue that generative systems "lack the kinds of experiences and cultural inheritances that structure every creative act" (Hall, 2024, p. 203). From this view, algorithmic generation imitates creativity without the consciousness that gives it meaning. Emerging newsroom practices—where reporters prompt models to draft text and then edit or curate results—illustrate a hybrid dance between human imagination and algorithmic prediction. Creativity thus becomes distributed across human and machine agents, redefining originality as relational rather than individual.



Together these oppositions demonstrate that algorithmic thinking does not simply replace the human in journalism but reconfigures the field's epistemic foundations. The challenge is to integrate AI's efficiency and scale without losing the intentionality, subjectivity, and lived creativity that constitute journalism's human essence.

Implications & Conclusion

The theoretical analysis suggests journalism is undergoing an epistemic shift toward hybrid cognition, where human judgment and algorithmic processes jointly produce news. Authorship evolves from solitary reporters to distributed creativity, raising questions of accountability and intellectual property — who is responsible when AI is involved? The concept of algorithmic subjectivity also emerges: AI systems, through their code and data choices, contribute implicit perspectives to news framing. To preserve trust, journalism must adopt algorithmic transparency and AI literacy for both producers and audiences. Researchers can operationalize these ideas by empirically comparing AI-assisted vs. human news, exploring newsroom workflows, and devising ethical frameworks. Practically, newsrooms should pilot hybrid models, set usage guidelines, and train teams in human-AI collaboration.

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ZAMONAVIY JURNALISTIKADA ANTROPOTSENTRIK DAN ALGORITMIK TAFAKKURGA O'TISH: IJODIY VA KOGNITIV PARADIGMALAR TRANSFORMATSIYASI

Annotatsiya. Ushbu tezisdagi zamonaviy jurnalistikada ijodiy va kognitiv paradigmlar antropotsentrik tafakkurdan algoritmik tafakkurga o'tish jarayonida qanday o'zgarayotgani tahlil qilinadi. Sun'iy intellekt, xususan, generativ AI tizimlari jurnalistik ijod, mualliflik va bilish jarayonlarini qayta shakllantirib, ayrim kognitiv funksiyalarni insondan mashinaga o'tkazmoqda. Maqolada jarayon falsafasi, postgumanizm va media ontologiyasi yondashuvlariga tayangan holda algoritmik jurnalistika insoniy niyat va mashinaviy optimallashtirish uyg'unlashgan gibrid tafakkur shakli sifatida talqin etiladi. Tadqiqotda bu jarayon jurnalistik ijodda asl g'oya, ishonchlilik va axloqiy mas'uliyat tushunchalariga qanday yangi ma'no baxsh etayotgani yoritiladi. Natijada, jurnalistika inson-mashina hamkorligiga asoslangan gibrid ijod va kollektiv mualliflik davriga qadam qo'yayotgani ta'kidlanadi.

Kalit so'zlar: algoritmik jurnalistika, ijodkorlik, mualliflik, postgumanizm, gibrid tafakkur, sun'iy intellekt, media epistemologiyasi



ПЕРЕХОД ОТ АНТРОПОЦЕНТРИЧЕСКОГО К АЛГОРИТМИЧЕСКОМУ МЫШЛЕНИЮ В СОВРЕМЕННОЙ ЖУРНАЛИСТИКЕ: ТРАНСФОРМАЦИЯ ТВОРЧЕСКИХ И КОГНИТИВНЫХ ПАРАДИГМ

Аннотация. В тезисе рассматривается трансформация творческих и когнитивных парадигм современной журналистики в условиях перехода от антропоцентрического к алгоритмическому мышлению. Показано, как системы генеративного искусственного интеллекта (ИИ) изменяют журналистское творчество, авторство и познание, перенося часть когнитивных функций от человека к машине. Опираясь на философию процесса, постгуманистическую теорию и медиа-онтологию, автор рассматривает алгоритмическую журналистику как гибридную форму мышления, сочетающую человеческое намерение и машинную оптимизацию. В работе также обсуждаются вызовы, связанные с оригинальностью, достоверностью и этической ответственностью в условиях автоматизации медиа. Делается вывод, что современная журналистика вступает в эпоху распределённого творчества и коллективного авторства, основанных на сотрудничестве человека и машины.

Ключевые слова: алгоритмическая журналистика, творчество, авторство, постгуманизм, гибридное мышление, искусственный интеллект, эпистемология медиа